Long-Context Pretraining

Deep Dive into Kimi K3 — Chapter 15

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Foundation Books Project

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Goal: type the record, verify its support, pack without leaking, account mixtures in the right unit, extend through gated stages, and report effective length per protocol.

Roadmap

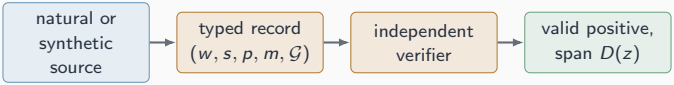
Records that carry dependencies

Packing, mixtures, and budgets

Stages, mechanisms, and diagnosis

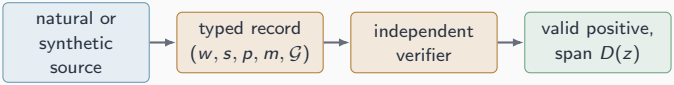
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Container length versus declared support



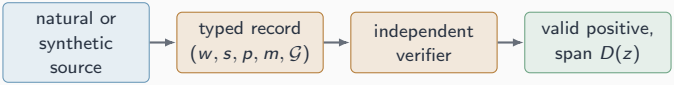
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- Label permutation must change the answer; a distractor permutation must not; a missing hop makes the positive unanswerable.
- K3 reports a 1M window, a long-data pipeline, and synthetic scattered-evidence tasks; thresholds and weights are undisclosed. [REPORTED: K3 Sec. 3.4]

Packing, mixtures, and budgets

Boundaries and accounting units

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- Equal-token stages: length 64 has one eighth the records of length 8 and over seven times the logical pairs — pairs are not FLOPs.

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Stages, mechanisms, and diagnosis

Gated extension and effective length

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Diagnose in dependency order. Next: Chapter 16, Native Multimodal Pretraining.